Answer key: Lesson 3, Digit Cards

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. You have two digit cards: 3 and 5. Make two numbers with both cards.

Answer: 35 and 53. 53 is greater.

With 3 on the left the number is 3 tens and 5 ones: 35. With 5 on the left it is 5 tens and 3 ones: 53. The same two cards make two numbers. 53 is greater, because 5 tens is more than 3 tens.

Watch for: Writing 8 (3 + 5).; Writing 35 twice.

S2. Write the greater number each time.

Answer: 72, 48, 65, 30

(a) 7 tens is more than 2 tens: 72. (b) Same tens, 8 ones is more than 1 one: 48. (c) 6 tens is more than 5 tens: 65. (d) 3 tens is more than 1 ten: 30.

Watch for: Choosing the number with the greater ones digit (saying 13 is greater than 30).; Not looking at the tens first.

S3. Look at the picture for Mei. Ben has 4 bars and 1 square.

Answer: Mei 38, Ben 41, Ben has more

Mei has 3 bars and 8 squares: 38. Ben has 4 bars and 1 square: 41. 4 tens is more than 3 tens, so Ben has the greater number, even though Mei has more squares.

Watch for: Saying Mei has more because she has 8 squares.; Reading Mei's picture as 83.

S4. Use the cards 6 and 2.

Answer: 62 and 26

The greatest card goes on the left for the greatest number: 62. The smallest card goes on the left for the smallest number: 26.

Watch for: Writing 62 for both.; Writing 8 (6 + 2).

S5. Arrange these in order. Begin with the smallest number: 52, 25, 41, 14.

Answer: 14, 25, 41, 52

14 has 1 ten, 25 has 2 tens, 41 has 4 tens, 52 has 5 tens. Smallest first: 14, 25, 41, 52.

Watch for: Putting 25 after 41 because 5 is greater than 1.; Writing them greatest first.

S6. Ravi says that 19 is greater than 21.

Answer: No. 21 is greater. 2 tens is more than 1 ten.

19 is 1 ten and 9 ones. 21 is 2 tens and 1 one. 2 tens is more than 1 ten, so 21 is greater. Ravi looked at the ones digits first. Look at the tens first.

Watch for: Saying Ravi is right because 9 is more than 2.; Not giving a reason (any clear sentence about the tens is fine).

S7. You have the cards 2 and 8. Use both cards each time.

Answer: 82 and 28

82 has 8 tens, which is more than 5 tens. 28 has only 2 tens, which is less than 5 tens.

Watch for: Writing 82 for both parts.; Writing 8 and 2.

S8. Siti has 25 stickers. Ravi has 52 stickers. Draw bars and squares in the box.

Answer: Ravi (52 is 5 tens and 2 ones; 25 is 2 tens and 5 ones)

Siti: 2 bars and 5 squares. Ravi: 5 bars and 2 squares. 5 bars is more than 2 bars, so Ravi has more stickers, even though both numbers use a 2 and a 5.

Watch for: Saying they have the same because they use the same digits.; Drawing 2 and 5 single squares for Siti.

Think harder (page 3)

T1. You have the cards 3, 5 and 8. Use two cards to make the greatest number. Then make the smallest number.

Answer: 85 and 35

Numbers you can make: 35, 38, 53, 58, 83, 85. The greatest is 85 (8 tens, the greatest card on the left, then the next greatest card). The smallest is 35.

Watch for: Using all three cards.; Writing 88 (using the 8 twice).

T2. You have the cards 0, 3 and 7. Use two cards to make the smallest and the greatest two-digit numbers.

Answer: 30 and 73

The 0 can not go on the left (03 is just 3). The two-digit numbers are 30, 37, 70 and 73. The smallest is 30 and the greatest is 73.

Watch for: Writing 03 or 3 for the smallest.; Writing 37 for the smallest.

T3. You have the cards 1, 3, 5 and 7. Make two-digit numbers with two different cards. The number is more than 40. Its left digit is greater than its right digit. Write all the numbers that fit.

Answer: 51, 53, 71, 73, 75

More than 40 means a tens card of 5 or 7. With 5 on the left: 51, 53, 57. With 7 on the left: 71, 73, 75. Clue 2 removes 57, because 5 is not greater than 7. Left: 51, 53, 71, 73, 75.

Watch for: Stopping after one or two numbers.; Writing 57 (5 is less than 7).

T4. You have the cards 2, 4 and 6. Use two different cards each time. Write all the two-digit numbers you can make.

Answer: 6 numbers: 24, 26, 42, 46, 62, 64

2 on the left: 24, 26. 4 on the left: 42, 46. 6 on the left: 62, 64. That is 2 + 2 + 2 = 6 numbers.

Watch for: Missing some by listing in a random order.; Writing 22, 44 or 66 (each card can be used only once).

Go further (page 4)

G1. You have the cards 4, 0 and 9. Use all three cards to make 3-digit numbers. Make the greatest number. Then make the smallest number.

Answer: 940 and 409

The ways are 409, 490, 904, 940 (049 and 094 start with 0, so they are not 3-digit numbers). The greatest is 940. The smallest is 409.

Watch for: Writing 049 as the smallest.; Writing 904 as the greatest (the 9 is on the left, but 4 should come next).

G2. You have the cards 1, 2, 3 and 4. Make two-digit numbers. How many numbers can you make?

Answer: 12 and 16

Each card once: 4 choices for the tens card, then 3 choices for the ones card, so 4 x 3 = 12. A card may be used twice: 4 choices and 4 choices, so 16 (the 12 and the four numbers 11, 22, 33, 44).

Watch for: Counting only 8 or 6 numbers.; Forgetting 11, 22, 33 and 44 in part b.

G3. I make a 3-digit number with the cards 2, 5 and 8. Each card is used once. My number is greater than 500 and less than 600. My ones digit is greater than my tens digit. What number am I?

Answer: 528

The numbers are 258, 285, 528, 582, 825, 852. More than 500: 528, 582, 825, 852. Less than 600: 528, 582. Ones digit greater than tens digit: 528 (8 is greater than 2). 582 has 2 and 8 the wrong way round.

Watch for: Stopping at 582 and not checking clue 2.; Listing only some of the 6 numbers.

Investigation (page 5): Is the greater card always first?

For two different cards, the greater card on the left always makes the greater number, because the tens decide. With the cards 0 and 8 you can only make 80, because 08 is just 8. With the same card twice (4 and 4) you can only make 44. So you can make two different two-digit numbers only when the two cards are different and neither is 0.

Look for: A child who tests several pairs, notices that the tens digit decides, and finds the two special cases (a 0 card, two matching cards). A reason using tens and ones is better than "because it is greater".

Exit ticket (page 5)

E1. Write the greater number: 39 or 41.

Answer: 41

41 has 4 tens and 39 has only 3 tens, so 41 is greater. Look at the tens first.

Watch for: Writing 39 because 9 is greater than 1.

E2. Use the cards 7 and 3.

Answer: 73 and 37

The greatest card on the left makes the greatest number: 73. The smallest card on the left makes the smallest number: 37.

Watch for: Writing 73 for both.; Writing 10 (7 + 3).

E3. Arrange these in order. Begin with the smallest number: 63, 36, 60.

Answer: 36, 60, 63

36 has 3 tens. 60 and 63 both have 6 tens. 60 has 0 ones and 63 has 3 ones, so 60 comes before 63.

Watch for: Putting 60 before 36 or after 63.; Writing them greatest first.

Sneak-peek puzzles (not in the workbook)

1. 18 stars.
2. Mei. Her number is 84, which is 8 tens. Ali has 48, which is only 4 tens.